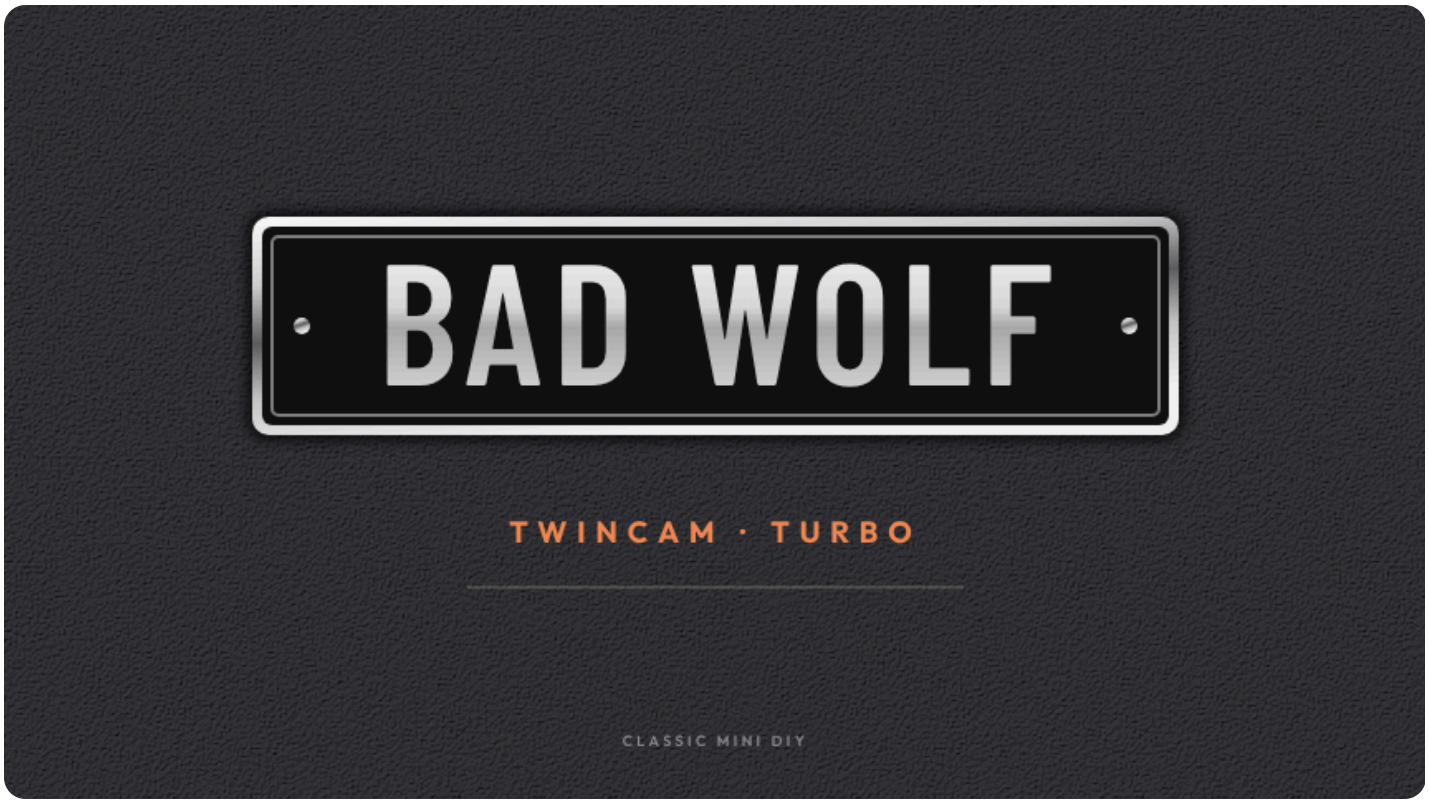
CLASSIC MINI DIY · HALTECH IC-7 · NSP

Bad Wolf iC-7 Dash Install Guide

Everything you need to put the Bad Wolf screens on your iC-7: the images, needles, lamps and fonts, plus where each piece goes. You build each screen once in NSP; after that it lives in your dash map like any Haltech screen.



What's in this guide

1. How custom screens work in NSP (and why the gauges are "empty")
2. Before you start: back up, versions, which five screens
3. What's in the bundle
4. Fonts and splash screen
5. The build method, once, in detail (with needle calibration)
6. One page per screen: every component, position and setting
7. Empty versions: blank layouts you can fill with any channel, any time
8. Write to the dash, day/night, troubleshooting

PART 1

How custom screens work in NSP

Since NSP 2.30, the iC-7 screens are fully editable. A screen is a stack of **components** you add with right-click › **Add Component**: **Images**, **Numeric Gauges**, **Dial Gauges**, bars and **Indicators**. Each has X, Y and Z position, a size, a channel and colours. Z is the layer: 0 is the back.

You were right that the gauges need to be "empty". Here is how this bundle does it:

1. **Everything that never moves is one picture.** The faces, numbers, ticks, chrome, tiles and labels are all printed into one 860 × 480 **background image**, placed as an Image component on the bottom layer.
2. **NSP draws only the moving parts on top.** Each needle is a Dial Gauge with its own dial made invisible (every palette colour set to opacity 0) and our needle image loaded as its **Needle Image Source**. Numbers are Numeric Gauges with only the value showing.
3. **Bars, lamps and status pills go underneath.** Wherever one lives, the background has a see-through cut-out: a rounded slot for a bar, a round lens or an arrow for a lamp, a pill shape for a status light (its label stays printed on top). A plain NSP component sits below the background and shows through, so the artwork gives it its shape and the component stays generic: a bar, or an indicator that is just a block of colour switching between its off and on colours.

This is the same method Haltech shows in its "Import Custom Screens and Needles" video: needle PNGs with a transparent background, **pointing straight up**, set with Needle Image Source and Needle Image Offset; and a dial gauge's palette opacities set to zero so only the needle shows.

Two versions of every screen. Printed (Parts 5 and 6): labels, units and scales are part of the artwork, so you only add needles and numbers. It's the quickest build and looks exactly like the previews, but changing what a gauge shows means new artwork. **Empty** (Part 7): only the hardware is in the picture (chrome, blank faces, plates, windows, tiles). NSP draws the ticks, numbers, labels and units, so you can swap any slot to any channel later without a redesign.

Screen size. The iC-7 layout canvas in NSP is **860 × 480**. Your own Bad Wolf dash map uses an 860 × 480 layout and splash, so every file here is made at exactly that size. Coordinates in this guide are screen pixels from the top-left corner.

Honest note. Haltech hasn't published written specs for the editor. Field names below are taken from Haltech's own tutorial videos, and they may differ slightly in your NSP version. Where something can't be known in advance (mainly how NSP scales a needle image), Part 5 has a one-time, two-minute calibration.

PART 2

Before you start

1. **Back up your current dash map.** Go online with the dash and save the dash map to a new file (your last one in the repo is `badwolf.ic7-2-34-0.nexmap`). If anything goes sideways, write that back.
2. **NSP 2.30 or newer** on the PC, and the dash updated to match. Your map is already 2.34, so you're set.
3. **Copy this whole folder to the Windows machine that runs NSP** (or open it from the VM's shared folder). NSP will ask for each file as you go.
4. **Pick five screens.** The dash holds up to five. A good set: **A Classic** · **B Tuning** (or C Six-pack / D Hybrid) · **E Diagnostic** · **F Diagnostic Dense** · plus your favourite of the rest.
5. **Channel names.** Channels below come from your October 2026 IO report. NSP's channel picker may word them a little differently; pick the closest match.

Tip. Build one screen on its own first (Classic is a good start), write it to the dash, and check it in the car before doing the rest. Once the first needle is calibrated, the rest go quickly.

PART 3

What's in the bundle

```
Bad Wolf iC-7 Dash/
├─ Install Guide - Bad Wolf iC-7 Dash.pdf      ← this file
├─ 1 - Fonts/                Outfit + Barlow Condensed (.ttf), licence note
├─ 2 - Splash Screen/        Plate (recommended), Dial, Car · 860×480
├─ 3 - Screens/
│  ├─ A - Classic/           classic_background_day/night, needles, lamps, classic_components.csv
│  ├─ B - Tuning/            AFR + target needle, oil T, fuel P, volts, air T
│  ├─ C - Six-pack/          six equal gauges
│  ├─ D - Hybrid/            big AFR + six data plates
│  ├─ E - Diagnostic/        diagnostic_background_dark/light, indicator fill
│  └─ F - Diagnostic Dense/  diagnostic_dense_background_dark/light, indicator fill
│     each folder also has "Previews (not for import)"
│     and "Empty version/": blank background(s) + …_slots_empty.csv
│     every file is named after its screen: tuning_needle_large_day.png…
└─ 4 - Reference/            layout spec (JSON), alternate needles, generator source
```

Every screen folder holds only what that screen needs. Files ending **_Day / _Night** (or **_Dark / _Light**) are the two looks; build one, or both as separate screens. Lamp files come in colour and as **_white** masks (see Part 5, Indicators).

PART 4

Fonts and splash screen

1 Fonts

NSP lets you upload your own font from a text component's **Font Source**. The first time a component needs Outfit or Barlow Condensed, choose **upload** and pick the .ttf from **1 - Fonts**; after that it's in the list.

- **Outfit SemiBold**: values on the Diagnostic screens.
- **Outfit Medium**: small secondary values (TGT, PW, PEDAL...).
- **Barlow Condensed SemiBold**: odometer, LCD readouts and Hybrid data plates.

If you'd also like them on the PC for previews, double-click each .ttf and choose Install. That's optional.

2 Splash screen

In the NSP tree under your iC-7, open **Splash Screen** and set its image to `splash_bad_wolf_plate.png` (recommended; or the dial or car version). They are 860 × 480, the same size as your current hero image.

PART 5

The build method

Every screen is built the same way. Do this once slowly; the per-screen tables then only give positions and settings.

1 Add a blank screen

In the tree go to **Screens** › **+ Add Screen** › **Blank Screen**. Rename it (for example "Classic").

2 Background (Z 1)

Right-click the empty screen › **Add Component** › **Images** › Image. Set **Image Source** to the screen's …`_background_….png`. Set X 0, Y 0, X size 860, Y size 480, Z 1. The bars go underneath it (Z 0); everything else sits on top.

3 Needles: a Dial Gauge with an invisible dial

1. Add Component › **Dial Gauges** › the retro circular gauge.
2. Type X, Y, X size and Y size from the table. The box is a square whose centre is the gauge's pivot.
3. **Channel**: from the table. **Min / Max**: from the table (Display Min / Display Max).
4. **Gauge Start Angle −135** and **Gauge End Angle 135** for every gauge (0° is straight up, so −90 is flat left and +90 is flat right).
5. **Needle Image Source** › Choose File › the …`_needle_….png` from the folder. Each one is a tight, narrow strip pointing straight up, with no empty space around it.
6. **Palette Colours**: set the opacity of every colour to 0 (numbers, ticks, arc, warning, centre, background) **except the Needle**. Turn tick counts to 0 if your version shows them anyway.

7. **Needle Image Offset −80%.** Every needle's pivot sits exactly 80% of the way down its image, above a short counterweight tail.
8. Add the **hub** (`…_hub_….png`) as an Image component on top (Z 5), centred on the pivot (positions in the table). It's round, so it never needs to turn.

Why the needles changed, and checking the first one. NSP sizes and places a needle image from its own outline. The first version of these needles was drawn on large square images with lots of empty space, so they could never be offset to pivot properly. The needles are now tight strips (for example 14 × 212 px for the large needle), with the boss moved to a separate hub image.

- Turn on **Simulation** and sweep the value. With −80% the needle should turn about the hub. If it's still a little off, nudge the offset by 1–2%.
- If your NSP measures the offset differently and −80% isn't close, try the **no-tail** needles in `4 - Reference/Alternate needles`. Their pivot is exactly on the bottom edge, so the offset is 0%.
- Length: if the tip falls short of or past the ticks, tick **Override Scaling** if your version shows it, or change the gauge's X/Y size equally, keeping the centre.

4 Numbers: Numeric Gauge, value only

Add Component › **Numeric Gauges**. Set position, size and channel from the table. Hide the label and units text (opacity 0, or delete the text), then set the value's font, size, colour and alignment as listed. Units are already printed on the background.

5 Bars

Bars go **behind** the background. Every place a bar lives has a see-through slot cut into the background image, with rounded ends. Put NSP's horizontal bar component under it (Z 0), at the position in the table or a few pixels larger on every side. The background then masks it, so the bar takes the slot's exact shape and the bar itself can stay plain and generic.

- **Fill colour**: as listed (olive on the diagnostic screens, ink or orange on the classic windows and plates).
- **Background colour**: the empty-track colour listed. It shows through the part of the slot the fill hasn't reached.
- Frame, border and any text on the bar: opacity 0.

A slot with no bar behind it shows the screen's black, so give every slot a bar, or a plain bar fixed at 0 as a filler.

6 Indicators: lamps and status pills (behind, like the bars)

Lamps and pills work exactly like the bars. Each lamp (round lens or turn arrow) and each status pill is a see-through cut-out in the background. The pill labels are printed on the background, on top.

- Add Component › **Indicator** at Z 0 (under the background), sized to the box in the table, which is the cut-out plus a small margin.
- **Image Source**: `…_indicator_fill_white.png`, a plain white square, so the indicator is just a block of colour. (If your NSP's indicator can show a plain colour without an image, that works too.)

- **Off colour**: the unlit colour from the table (dark lens, faint arrow, dim pill). **On colour**: the lamp or status colour. NSP switches between them and the cut-out shapes the result.
- Every cut-out needs something behind it, or it shows the screen's black.

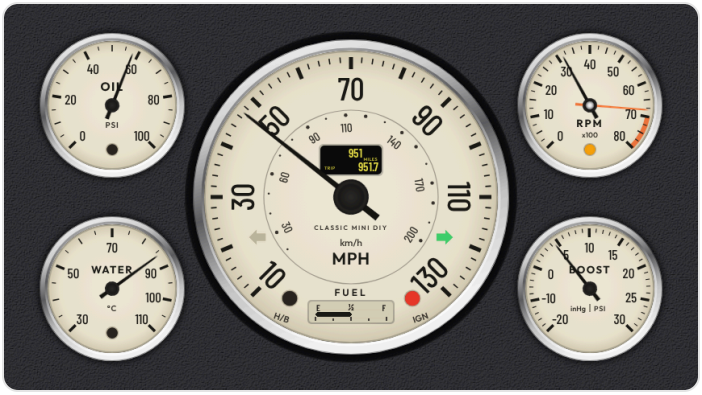
7 Layers

Z 0: bars, lamps and pills (under the background) · **Z 1: background** · Z 2: numbers · Z 3: secondary needles · Z 4: main needles · Z 5: hubs. Higher numbers draw on top.

SCREEN A · FOLDER “3 - SCREENS/A - CLASSIC”

Classic

The HMW pod redrawn for the iC-7: speedo with odometer and fuel window, oil, water, tacho and boost.



Day (preview with sample values)



Night

Build it top to bottom. X/Y are the top-left corner in screen pixels; “X size/Y size” are the component's width and height. The same table is in the folder's `…_components.csv`. Every file starts with the screen's name (for example `diagnostic_background_dark.png`).

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
1	**Background**	Image	`classic_background_day.png (or classic_background_night.png)`	0	0	860	480	1	—	Fill the whole screen. Above the bars, below everything else. Its see-through slots show the bars underneath.
2	**HighBeam lamp**	Indicator	`classic_indicator_fill_white.png`	342	354	24	24	0	Head Light Input (PD16 AVI2) or your high-beam channel	Behind background (round lens) · On #4F8FF7 · Off #2A2620 day / #0D0C0B night
3	**Ignition lamp**	Indicator	`classic_indicator_fill_white.png`	494	354	24	24	0	Battery Voltage Light (PD16 HBO1)	Behind background (round lens) · On #E8382C · Off #2A2620 day / #0D0C0B night
4	**TurnLeft lamp**	Indicator	`classic_indicator_fill_white.png`	301	277	26	26	0	Left Indicator Output (PD16 8A-HCO6)	Behind background (arrow) · On #3FCF6A · Off #BDB49C day / #45413A night
5	**TurnRight lamp**	Indicator	`classic_indicator_fill_white.png`	533	277	26	26	0	Right Indicator Output (PD16 8A-HCO2)	Behind background (arrow) · On #3FCF6A · Off #BDB49C day / #45413A night
6	**OilPressure lamp**	Indicator	`classic_indicator_fill_white.png`	124	172	19	19	0	Oil Pressure Light (DPO3)	Behind background (round lens) · On #E8382C · Off #2A2620 day / #0D0C0B night
7	**Fan lamp**	Indicator	`classic_indicator_fill_white.png`	124	400	19	19	0	Thermofan 1 (25A-HCO4)	Behind background (round lens) · On #A9BF86 · Off #2A2620 day / #0D0C0B night
8	**CheckEngine lamp**	Indicator	`classic_indicator_fill_white.png`	716	172	19	19	0	Check Engine Light (DPO5)	Behind background (round lens) · On #F5A00B · Off #2A2620 day / #0D0C0B night
9	**Odometer**	Numeric gauge		396	177	48	17	2	Odometer	Value only · Barlow Condensed SemiBold 15px · right-aligned · #E9E04A day · #C9C03C night
10	**Trip**	Numeric gauge		416	194	48	17	2	Trip Meter	Value only · Barlow Condensed SemiBold 15px · right-aligned · #E9E04A day · #C9C03C night

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
11	**Fuel level bar**	Bar gauge (horizontal)		386	383	88	5	0	Fuel Level 1 (PD16 SPI4)	Behind background · Min 0 / Max 100 · fill #1D1B18 day · #F08A54 night · track #D6CDB4 day · #1D1C19 night · frame off
12	**Tacho peak marker (optional)**	Dial gauge (Retro circular)	`classic_needle_peakrpm_day/night.png`	644	44	164	164	3	Peak / max RPM channel, if your firmware has one	Min 0 / Max 8000 rpm · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (726, 126)
13	**Speedometer**	Dial gauge (Retro circular)	`classic_needle_large_day/night.png`	252	62	356	356	4	Vehicle Speed (SPI1 drive train)	Min 10 / Max 130 mph · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (430, 240) · Face starts at 10 mph: to let the needle rest below 10, use Min 0 and Start −157.5°
14	**Tachometer**	Dial gauge (Retro circular)	`classic_needle_tacho_day/night.png`	648	48	156	156	4	Engine Speed (RPM)	Min 0 / Max 8000 rpm · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (726, 126)
15	**Oil pressure**	Dial gauge (Retro circular)	`classic_needle_small_day/night.png`	56	48	156	156	4	Oil Pressure (AVI4)	Min 0 / Max 100 psi · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (134, 126)
16	**Water temp**	Dial gauge (Retro circular)	`classic_needle_small_day/night.png`	56	276	156	156	4	Coolant Temperature (AVI3)	Min 30 / Max 110 °C · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (134, 354)
17	**Boost**	Dial gauge (Retro circular)	`classic_needle_small_day/night.png`	648	276	156	156	4	Manifold Pressure (OBPS 1) shown as gauge pressure	Min -9.82 / Max 30 psi (gauge) · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (726, 354) · Linear −20 inHg (−9.82 psi) to +30 psi; in kPa(g): −67.7 / 206.8
18	**Speedometer hub**	Image	`classic_hub_large.png`	406	216	48	48	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
19	**Tachometer hub**	Image	`classic_hub_tacho.png`	715	115	22	22	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
20	**Oil pressure hub**	Image	`classic_hub_small.png`	123	115	22	22	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
21	**Water temp hub**	Image	`classic_hub_small.png`	123	343	22	22	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
22	**Boost hub**	Image	`classic_hub_small.png`	715	343	22	22	5	—	Round centre cap on top of the needle at the pivot. Never rotates.

SCREEN B · FOLDER “3 - SCREENS/B - TUNING”

Classic · Tuning

What the analog gauges don't show: a big air/fuel gauge with lambda ring and target needle, oil temp, fuel pressure, volts and intake air, plus throttle position, knock and logging lamps.



Day (preview with sample values)



Night

Build it top to bottom. X/Y are the top-left corner in screen pixels; “X size/Y size” are the component's width and height. The same table is in the folder's …_components.csv. Every file starts with the screen's name (for example diagnostic_background_dark.png).

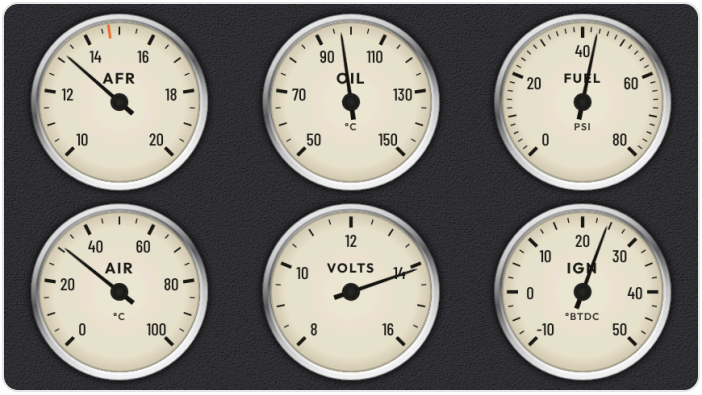
#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
1	**Background**	Image	tuning_background_day.png (or tuning_background_night.png)	0	0	860	480	1	—	Fill the whole screen. Above the bars, below everything else. Its see-through slots show the bars underneath.
2	**Knock lamp**	Indicator	tuning_indicator_fill_white.png	342	354	24	24	0	Knock Level above threshold	Behind background (round lens) · On #F5A00B · Off #2A2620 day / #0D0C0B night
3	**Logging lamp**	Indicator	tuning_indicator_fill_white.png	494	354	24	24	0	Datalog Switch (Keypad B5) / logging active	Behind background (round lens) · On #3FCF6A · Off #2A2620 day / #0D0C0B night
4	**AFR (LCD)**	Numeric gauge		396	177	68	17	2	Wideband O2 1 (internal WBO), as AFR	Value only · Barlow Condensed SemiBold 15px · right-aligned · #E9E04A day · #C9C03C night
5	**AFR target (LCD)**	Numeric gauge		396	194	68	17	2	Target AFR	Value only · Barlow Condensed SemiBold 15px · right-aligned · #E9E04A day · #C9C03C night
6	**Throttle bar**	Bar gauge (horizontal)		386	383	88	5	0	Throttle Position (DBW TPS 1)	Behind background · Min 0 / Max 100 · fill #1D1B18 day · #F08A54 night · track #D6CDB4 day · #1D1C19 night · frame off
7	**Air / fuel**	Dial gauge (Retro circular)	tuning_needle_large_day/night.png	252	62	356	356	4	Wideband O2 1 (internal WBO), as AFR	Min 10 / Max 20 AFR · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (430, 240)
8	**AFR target needle**	Dial gauge (Retro circular)	tuning_needle_afrtarget_day/night.png	248	58	364	364	3	Target AFR	Min 10 / Max 20 AFR · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (430, 240) · Sits under the main AFR needle
9	**Oil temp**	Dial gauge (Retro circular)	tuning_needle_small_day/night.png	56	48	156	156	4	Oil Temperature (AVI10)	Min 50 / Max 150 °C · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (134, 126)
10	**Fuel pressure**	Dial gauge (Retro circular)	tuning_needle_small_day/night.png	56	276	156	156	4	Fuel Pressure (AVI11)	Min 0 / Max 80 psi · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (134, 354)

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
11	**Volts**	Dial gauge (Retro circular)	`tuning_needle_small_day/night.png`	648	48	156	156	4	ECU Supply Voltage	Min 8 / Max 16 V · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (726, 126)
12	**Intake air**	Dial gauge (Retro circular)	`tuning_needle_small_day/night.png`	648	276	156	156	4	Intake Air Temperature (AVI7)	Min 0 / Max 100 °C · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (726, 354)
13	**Air / fuel hub**	Image	`tuning_hub_large.png`	406	216	48	48	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
14	**Oil temp hub**	Image	`tuning_hub_small.png`	123	115	22	22	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
15	**Fuel pressure hub**	Image	`tuning_hub_small.png`	123	343	22	22	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
16	**Volts hub**	Image	`tuning_hub_small.png`	715	115	22	22	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
17	**Intake air hub**	Image	`tuning_hub_small.png`	715	343	22	22	5	—	Round centre cap on top of the needle at the pivot. Never rotates.

SCREEN C · FOLDER “3 - SCREENS/C - SIX-PACK”

Classic · Six-pack

Six equal gauges, the easiest to read at a glance: AFR, oil temp, fuel pressure, intake air, volts and ignition timing.



Day (preview with sample values)



Night

Build it top to bottom. X/Y are the top-left corner in screen pixels; “X size/Y size” are the component's width and height. The same table is in the folder's …_components.csv. Every file starts with the screen's name (for example diagnostic_background_dark.png).

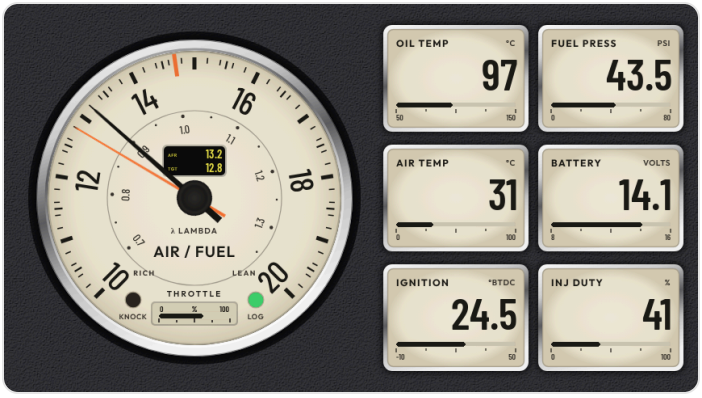
#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
1	**Background**	Image	sixpack_background_day.png (or sixpack_background_night.png)	0	0	860	480	1	—	Fill the whole screen. Above the bars, below everything else. Its see-through slots show the bars underneath.
2	**Air / fuel**	Dial gauge (Retro circular)	sixpack_needle_medium_day/night.png	47	26	192	192	4	Wideband O2 1 (internal WBO), as AFR	Min 10 / Max 20 AFR · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (143, 122)
3	**Oil temp**	Dial gauge (Retro circular)	sixpack_needle_medium_day/night.png	334	26	192	192	4	Oil Temperature (AVI10)	Min 50 / Max 150 °C · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (430, 122)
4	**Fuel pressure**	Dial gauge (Retro circular)	sixpack_needle_medium_day/night.png	621	26	192	192	4	Fuel Pressure (AVI11)	Min 0 / Max 80 psi · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (717, 122)
5	**Intake air**	Dial gauge (Retro circular)	sixpack_needle_medium_day/night.png	47	262	192	192	4	Intake Air Temperature (AVI7)	Min 0 / Max 100 °C · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (143, 358)
6	**Volts**	Dial gauge (Retro circular)	sixpack_needle_medium_day/night.png	334	262	192	192	4	ECU Supply Voltage	Min 8 / Max 16 V · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (430, 358)
7	**Ignition timing**	Dial gauge (Retro circular)	sixpack_needle_medium_day/night.png	621	262	192	192	4	Ignition Angle	Min -10 / Max 50 °BTDC · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (717, 358)
8	**Air / fuel hub**	Image	sixpack_hub_medium.png	130	109	26	26	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
9	**Oil temp hub**	Image	sixpack_hub_medium.png	417	109	26	26	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
10	**Fuel pressure hub**	Image	sixpack_hub_medium.png	704	109	26	26	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
11	**Intake air hub**	Image	sixpack_hub_medium.png	130	345	26	26	5	—	Round centre cap on top of the needle at the pivot. Never rotates.
12	**Volts hub**	Image	sixpack_hub_medium.png	417	345	26	26	5	—	Round centre cap on top of the needle at the pivot. Never rotates.

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
13	**Ignition timing hub**	Image	`sixpack_hub_medium.png`	704	345	26	26	5	—	Round centre cap on top of the needle at the pivot. Never rotates.

SCREEN D · FOLDER “3 - SCREENS/D - HYBRID”

Classic · Hybrid

The big AFR gauge plus six magnolia data plates with large numbers: oil temp, fuel pressure, air temp, battery, ignition and injector duty.



Day (preview with sample values)



Night

Build it top to bottom. X/Y are the top-left corner in screen pixels; “X size/Y size” are the component's width and height. The same table is in the folder's `…_components.csv`. Every file starts with the screen's name (for example `diagnostic_background_dark.png`).

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
1	**Background**	Image	`hybrid_background_day.png (or hybrid_background_night.png)`	0	0	860	480	1	—	Fill the whole screen. Above the bars, below everything else. Its see-through slots show the bars underneath.
2	**Knock lamp**	Indicator	`hybrid_indicator_fill_white.png`	148	354	24	24	0	Knock Level above threshold	Behind background (round lens) · On #F5A00B · Off #2A2620 day / #0D0C0B night
3	**Logging lamp**	Indicator	`hybrid_indicator_fill_white.png`	300	354	24	24	0	Datalog Switch (Keypad B5) / logging active	Behind background (round lens) · On #3FCF6A · Off #2A2620 day / #0D0C0B night
4	**AFR (LCD)**	Numeric gauge		202	177	68	17	2	Wideband O2 1 (internal WBO), as AFR	Value only · Barlow Condensed SemiBold 15px · right-aligned · #E9E04A day · #C9C03C night
5	**AFR target (LCD)**	Numeric gauge		202	194	68	17	2	Target AFR	Value only · Barlow Condensed SemiBold 15px · right-aligned · #E9E04A day · #C9C03C night
6	**Throttle bar**	Bar gauge (horizontal)		192	383	88	5	0	Throttle Position (DBW TPS 1)	Behind background · Min 0 / Max 100 · fill #1D1B18 day · #F08A54 night · track #D6CDB4 day · #1D1C19 night · frame off
7	**Oil temp value**	Numeric gauge		484	58	152	54	2	Oil Temperature (AVI10)	Value only · Barlow Condensed SemiBold 54px · right-aligned · #1d1b18 day · #e9e2cc night · 0 dp
8	**Oil temp bar**	Bar gauge (horizontal)		486	122	148	5	0	Oil Temperature (AVI10)	Behind background · Min 50 / Max 150 · fill #1d1b18 day · #f08a54 night · track #CCC5B3 day · #403D37 night · frame off
9	**Fuel press value**	Numeric gauge		676	58	152	54	2	Fuel Pressure (AVI11)	Value only · Barlow Condensed SemiBold 54px · right-aligned · #1d1b18 day · #e9e2cc night · 1 dp
10	**Fuel press bar**	Bar gauge (horizontal)		678	122	148	5	0	Fuel Pressure (AVI11)	Behind background · Min 0 / Max 80 · fill #1d1b18 day · #f08a54 night · track #CCC5B3 day · #403D37 night · frame off

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
11	**Air temp value**	Numeric gauge		484	206	152	54	2	Intake Air Temperature (AVI7)	Value only · Barlow Condensed SemiBold 54px · right-aligned · #1d1b18 day · #e9e2cc night · 0 dp
12	**Air temp bar**	Bar gauge (horizontal)		486	270	148	5	0	Intake Air Temperature (AVI7)	Behind background · Min 0 / Max 100 · fill #1d1b18 day · #f08a54 night · track #CCC5B3 day · #403D37 night · frame off
13	**Battery value**	Numeric gauge		676	206	152	54	2	ECU Supply Voltage	Value only · Barlow Condensed SemiBold 54px · right-aligned · #1d1b18 day · #e9e2cc night · 1 dp
14	**Battery bar**	Bar gauge (horizontal)		678	270	148	5	0	ECU Supply Voltage	Behind background · Min 8 / Max 16 · fill #1d1b18 day · #f08a54 night · track #CCC5B3 day · #403D37 night · frame off
15	**Ignition value**	Numeric gauge		484	354	152	54	2	Ignition Angle	Value only · Barlow Condensed SemiBold 54px · right-aligned · #1d1b18 day · #e9e2cc night · 1 dp
16	**Ignition bar**	Bar gauge (horizontal)		486	418	148	5	0	Ignition Angle	Behind background · Min -10 / Max 50 · fill #1d1b18 day · #f08a54 night · track #CCC5B3 day · #403D37 night · frame off
17	**Inj duty value**	Numeric gauge		676	354	152	54	2	Injector Duty Cycle (primary)	Value only · Barlow Condensed SemiBold 54px · right-aligned · #1d1b18 day · #e9e2cc night · 0 dp
18	**Inj duty bar**	Bar gauge (horizontal)		678	418	148	5	0	Injector Duty Cycle (primary)	Behind background · Min 0 / Max 100 · fill #1d1b18 day · #f08a54 night · track #CCC5B3 day · #403D37 night · frame off
19	**Air / fuel**	Dial gauge (Retro circular)	`hybrid_needle_large_day/night.png`	58	62	356	356	4	Wideband O2 1 (internal WBO), as AFR	Min 10 / Max 20 AFR · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (236, 240)
20	**AFR target needle**	Dial gauge (Retro circular)	`hybrid_needle_afrtarget_day/night.png`	54	58	364	364	3	Target AFR	Min 10 / Max 20 AFR · Start −135° · End 135° · Needle Image Offset −80% · palette off except Needle · Centre (236, 240) · Sits under the main AFR needle
21	**Air / fuel hub**	Image	`hybrid_hub_large.png`	212	216	48	48	5	—	Round centre cap on top of the needle at the pivot. Never rotates.

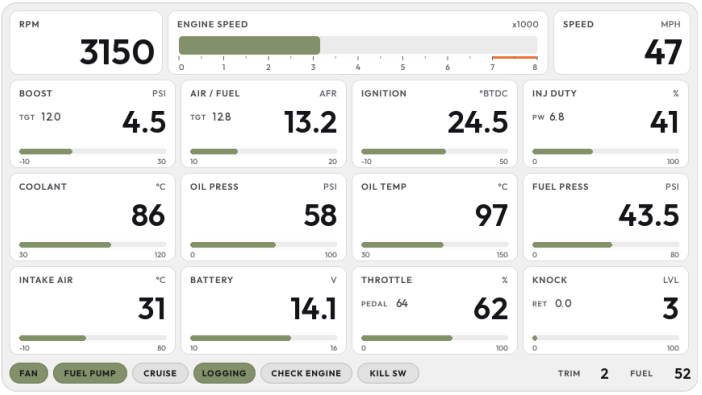
SCREEN E · FOLDER “3 - SCREENS/E - DIAGNOSTIC”

Diagnostic

Twelve tiles for dyno days and fault-finding, an RPM bar with redline, and status pills that light from behind.



Dark (preview with sample values)



Light

Build it top to bottom. X/Y are the top-left corner in screen pixels; “X size/Y size” are the component's width and height. The same table is in the folder's `…_components.csv`. Every file starts with the screen's name (for example `diagnostic_background_dark.png`).

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
1	**Background**	Image	`diagnostic_background_dark.png (or diagnostic_background_light.png)`	0	0	860	480	1	—	Fill the whole screen. Above the bars, below everything else. Its see-through slots show the bars underneath.
2	**rpm value**	Numeric gauge		18	36	170	46	2	Engine Speed (RPM)	Value only · Outfit SemiBold 44px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
3	**rpm bar**	Bar gauge (horizontal)		218	40	444	22	0	Engine Speed (RPM)	Behind background · Min 0 / Max 8000 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
4	**speed value**	Numeric gauge		692	36	150	46	2	Vehicle Speed (SPI1 drive train)	Value only · Outfit SemiBold 44px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
5	**boost value**	Numeric gauge		92	124	110	44	2	Manifold Pressure (gauge)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
6	**boost bar**	Bar gauge (horizontal)		20	180	182	6	0	Manifold Pressure (gauge)	Behind background · Min -10 / Max 30 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
7	**boost tgt**	Numeric gauge		48	132	42	16	2	Boost Target	Value only · Outfit Medium 13px · left-aligned · #b6b6bc dark · #4a4a4f light
8	**afr value**	Numeric gauge		304	124	110	44	2	Wideband O2 1 (internal WBO)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
9	**afr bar**	Bar gauge (horizontal)		232	180	182	6	0	Wideband O2 1 (internal WBO)	Behind background · Min 10 / Max 20 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
10	**afr tgt**	Numeric gauge		260	132	42	16	2	Target AFR / Lambda	Value only · Outfit Medium 13px · left-aligned · #b6b6bc dark · #4a4a4f light
11	**ign value**	Numeric gauge		444	124	182	44	2	Ignition Angle	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
12	**ign bar**	Bar gauge (horizontal)		444	180	182	6	0	Ignition Angle	Behind background · Min -10 / Max 50 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
13	**inj value**	Numeric gauge		728	124	110	44	2	Injector Duty Cycle (primary)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
14	**inj bar**	Bar gauge (horizontal)		656	180	182	6	0	Injector Duty Cycle (primary)	Behind background · Min 0 / Max 100 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
15	**inj pw**	Numeric gauge		677	132	49	16	2	Injection Time (ms)	Value only · Outfit Medium 13px · left-aligned · #b6b6bc dark · #4a4a4f light
16	**clt value**	Numeric gauge		20	240	182	44	2	Coolant Temperature (AVI3)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
17	**clt bar**	Bar gauge (horizontal)		20	296	182	6	0	Coolant Temperature (AVI3)	Behind background · Min 30 / Max 120 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
18	**oilp value**	Numeric gauge		232	240	182	44	2	Oil Pressure (AVI4)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
19	**oilp bar**	Bar gauge (horizontal)		232	296	182	6	0	Oil Pressure (AVI4)	Behind background · Min 0 / Max 100 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
20	**oilt value**	Numeric gauge		444	240	182	44	2	Oil Temperature (AVI10)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
21	**oilt bar**	Bar gauge (horizontal)		444	296	182	6	0	Oil Temperature (AVI10)	Behind background · Min 30 / Max 150 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
22	**fuelp value**	Numeric gauge		656	240	182	44	2	Fuel Pressure (AVI11)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
23	**fuelp bar**	Bar gauge (horizontal)		656	296	182	6	0	Fuel Pressure (AVI11)	Behind background · Min 0 / Max 80 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
24	**iat value**	Numeric gauge		20	356	182	44	2	Intake Air Temperature (AVI7)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
25	**iat bar**	Bar gauge (horizontal)		20	412	182	6	0	Intake Air Temperature (AVI7)	Behind background · Min -10 / Max 80 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
26	**batt value**	Numeric gauge		232	356	182	44	2	ECU Supply Voltage	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
27	**batt bar**	Bar gauge (horizontal)		232	412	182	6	0	ECU Supply Voltage	Behind background · Min 10 / Max 16 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
28	**tps value**	Numeric gauge		516	356	110	44	2	Throttle Position (DBW TPS 1)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
29	**tps bar**	Bar gauge (horizontal)		444	412	182	6	0	Throttle Position (DBW TPS 1)	Behind background · Min 0 / Max 100 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
30	**tps pedal**	Numeric gauge		487	364	28	16	2	Pedal Position (APS A/B)	Value only · Outfit Medium 13px · left-aligned · #b6b6bc dark · #4a4a4f light
31	**knock value**	Numeric gauge		728	356	110	44	2	Knock Level (KNOCK1)	Value only · Outfit SemiBold 38px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
32	**knock bar**	Bar gauge (horizontal)		656	412	182	6	0	Knock Level (KNOCK1)	Behind background · Min 0 / Max 100 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
33	**knock ret**	Numeric gauge		684	364	42	16	2	Knock Retard (°)	Value only · Outfit Medium 13px · left-aligned · #b6b6bc dark · #4a4a4f light
34	**FAN pill**	Indicator	`diagnostic_indicator_fill_white.png`	8	446	48	26	0	Thermofan 1 Output (25A-HCO4)	Behind background (pill; label printed) · On #9aac7f dark / #859369 light · Off #3A3A40 dark / #E4E4E4 light

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
35	**PUMP pill**	Indicator	`diagnostic_indicator_fill_white.png`	62	446	92	26	0	Primary Fuel Pump (PD16 25A-HCO3)	Behind background (pill; label printed) · On #9aac7f dark / #859369 light · Off #3A3A40 dark / #E4E4E4 light
36	**CRUISE pill**	Indicator	`diagnostic_indicator_fill_white.png`	160	446	70	26	0	Cruise Control Enable (Keypad B1)	Behind background (pill; label printed) · On #9aac7f dark / #859369 light · Off #3A3A40 dark / #E4E4E4 light
37	**LOG pill**	Indicator	`diagnostic_indicator_fill_white.png`	236	446	77	26	0	Datalog Switch (Keypad B5)	Behind background (pill; label printed) · On #9aac7f dark / #859369 light · Off #3A3A40 dark / #E4E4E4 light
38	**CEL pill**	Indicator	`diagnostic_indicator_fill_white.png`	319	446	114	26	0	Check Engine Light (DPO5)	Behind background (pill; label printed) · On #fbbf24 dark / #f59e0b light · Off #3A3A40 dark / #E4E4E4 light
39	**KILL pill**	Indicator	`diagnostic_indicator_fill_white.png`	439	446	77	26	0	Kill Switch Input 1 (GenCond5)	Behind background (pill; label printed) · On #ef7676 dark / #b22222 light · Off #3A3A40 dark / #E4E4E4 light
40	**fuel value**	Numeric gauge		810	448	42	22	2	Fuel Level 1 (PD16 SPI4)	Value only · Outfit SemiBold 18px · right-aligned · #e4e4e7 dark · #18181b light
41	**trim value**	Numeric gauge		720	448	30	22	2	Rotary Trim Module 1 (Boost Trim)	Value only · Outfit SemiBold 18px · right-aligned · #e4e4e7 dark · #18181b light

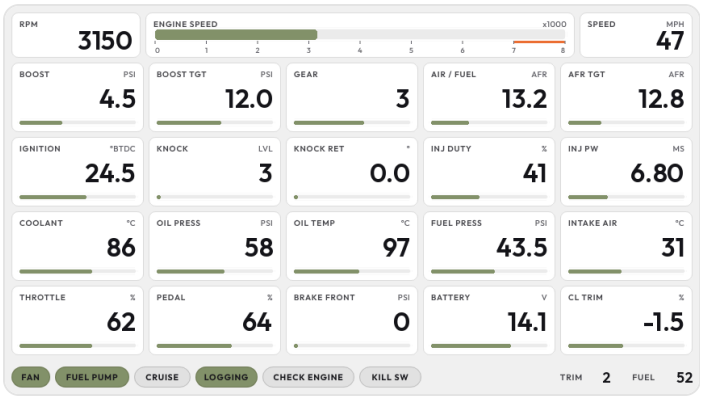
SCREEN F · FOLDER “3 - SCREENS/F - DIAGNOSTIC DENSE”

Diagnostic · Dense

Twenty tiles in four themed rows: boost and fuelling, spark and injection, temperatures and pressures, driver inputs and electrics.



Dark (preview with sample values)



Light

Build it top to bottom. X/Y are the top-left corner in screen pixels; “X size/Y size” are the component's width and height. The same table is in the folder's `…_components.csv`. Every file starts with the screen's name (for example `diagnostic_background_dark.png`).

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
1	**Background**	Image	`diagnostic_dense_background_dark.png (or diagnostic_dense_background_light.png)`	0	0	860	480	1	—	Fill the whole screen. Above the bars, below everything else. Its see-through slots show the bars underneath.
2	**rpm value**	Numeric gauge		18	24	140	36	2	Engine Speed (RPM)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
3	**rpm bar**	Bar gauge (horizontal)		186	29	508	12	0	Engine Speed (RPM)	Behind background · Min 0 / Max 8000 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
4	**speed value**	Numeric gauge		722	24	120	36	2	Vehicle Speed (SPI1 drive train)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
5	**boost value**	Numeric gauge		18	94	144	40	2	Manifold Pressure (gauge)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
6	**boost bar**	Bar gauge (horizontal)		18	142	144	4	0	Manifold Pressure (gauge)	Behind background · Min -10 / Max 30 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
7	**boost tgt value**	Numeric gauge		188	94	144	40	2	Boost Target	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
8	**boost tgt bar**	Bar gauge (horizontal)		188	142	144	4	0	Boost Target	Behind background · Min -10 / Max 30 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
9	**gear value**	Numeric gauge		358	94	144	40	2	Gear Position (calculated from speed / RPM)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
10	**gear bar**	Bar gauge (horizontal)		358	142	144	4	0	Gear Position (calculated from speed / RPM)	Behind background · Min 0 / Max 5 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
11	**afr value**	Numeric gauge		528	94	144	40	2	Wideband O2 1 (internal WBO)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
12	**afr bar**	Bar gauge (horizontal)		528	142	144	4	0	Wideband O2 1 (internal WBO)	Behind background · Min 10 / Max 20 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
13	**afr tgt value**	Numeric gauge		698	94	144	40	2	Target AFR	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
14	**afr tgt bar**	Bar gauge (horizontal)		698	142	144	4	0	Target AFR	Behind background · Min 10 / Max 20 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
15	**ign value**	Numeric gauge		18	186	144	40	2	Ignition Angle	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
16	**ign bar**	Bar gauge (horizontal)		18	234	144	4	0	Ignition Angle	Behind background · Min -10 / Max 50 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
17	**knock value**	Numeric gauge		188	186	144	40	2	Knock Level (KNOCK1)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
18	**knock bar**	Bar gauge (horizontal)		188	234	144	4	0	Knock Level (KNOCK1)	Behind background · Min 0 / Max 100 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
19	**knock ret value**	Numeric gauge		358	186	144	40	2	Knock Retard	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
20	**knock ret bar**	Bar gauge (horizontal)		358	234	144	4	0	Knock Retard	Behind background · Min 0 / Max 10 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
21	**inj value**	Numeric gauge		528	186	144	40	2	Injector Duty Cycle (primary)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
22	**inj bar**	Bar gauge (horizontal)		528	234	144	4	0	Injector Duty Cycle (primary)	Behind background · Min 0 / Max 100 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
23	**inj pw value**	Numeric gauge		698	186	144	40	2	Injection Time	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 2 dp
24	**inj pw bar**	Bar gauge (horizontal)		698	234	144	4	0	Injection Time	Behind background · Min 0 / Max 20 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
25	**clt value**	Numeric gauge		18	278	144	40	2	Coolant Temperature (AVI3)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
26	**clt bar**	Bar gauge (horizontal)		18	326	144	4	0	Coolant Temperature (AVI3)	Behind background · Min 30 / Max 120 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
27	**oilp value**	Numeric gauge		188	278	144	40	2	Oil Pressure (AVI4)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
28	**oilp bar**	Bar gauge (horizontal)		188	326	144	4	0	Oil Pressure (AVI4)	Behind background · Min 0 / Max 100 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
29	**oilt value**	Numeric gauge		358	278	144	40	2	Oil Temperature (AVI10)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
30	**oilt bar**	Bar gauge (horizontal)		358	326	144	4	0	Oil Temperature (AVI10)	Behind background · Min 30 / Max 150 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
31	**fuelp value**	Numeric gauge		528	278	144	40	2	Fuel Pressure (AVI11)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
32	**fuelp bar**	Bar gauge (horizontal)		528	326	144	4	0	Fuel Pressure (AVI11)	Behind background · Min 0 / Max 80 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off

#	COMPONENT	NSP TYPE	FILE	X	Y	X SIZE	Y SIZE	Z	CHANNEL	SETTINGS
33	**iat value**	Numeric gauge		698	278	144	40	2	Intake Air Temperature (AVI7)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
34	**iat bar**	Bar gauge (horizontal)		698	326	144	4	0	Intake Air Temperature (AVI7)	Behind background · Min -10 / Max 80 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
35	**tps value**	Numeric gauge		18	370	144	40	2	Throttle Position (DBW TPS 1)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
36	**tps bar**	Bar gauge (horizontal)		18	418	144	4	0	Throttle Position (DBW TPS 1)	Behind background · Min 0 / Max 100 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
37	**pedal value**	Numeric gauge		188	370	144	40	2	Pedal Position (APS A/B)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
38	**pedal bar**	Bar gauge (horizontal)		188	418	144	4	0	Pedal Position (APS A/B)	Behind background · Min 0 / Max 100 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
39	**brake value**	Numeric gauge		358	370	144	40	2	Brake Pressure Front (SPI3)	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 0 dp
40	**brake bar**	Bar gauge (horizontal)		358	418	144	4	0	Brake Pressure Front (SPI3)	Behind background · Min 0 / Max 1500 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
41	**batt value**	Numeric gauge		528	370	144	40	2	ECU Supply Voltage	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
42	**batt bar**	Bar gauge (horizontal)		528	418	144	4	0	ECU Supply Voltage	Behind background · Min 10 / Max 16 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
43	**cl trim value**	Numeric gauge		698	370	144	40	2	Closed Loop Fuel Correction	Value only · Outfit SemiBold 32px · right-aligned · #e4e4e7 dark · #18181b light · 1 dp
44	**cl trim bar**	Bar gauge (horizontal)		698	418	144	4	0	Closed Loop Fuel Correction	Behind background · Min -25 / Max 25 · fill #9aac7f dark · #859369 light · track #333337 dark · #EBEBEB light · frame off
45	**FAN pill**	Indicator	`diagnostic_dense_indicator_fill_white.png`	8	446	48	26	0	Thermofan 1 Output (25A-HCO4)	Behind background (pill; label printed) · On #9aac7f dark / #859369 light · Off #3A3A40 dark / #E4E4E4 light
46	**PUMP pill**	Indicator	`diagnostic_dense_indicator_fill_white.png`	62	446	92	26	0	Primary Fuel Pump (PD16 25A-HCO3)	Behind background (pill; label printed) · On #9aac7f dark / #859369 light · Off #3A3A40 dark / #E4E4E4 light
47	**CRUISE pill**	Indicator	`diagnostic_dense_indicator_fill_white.png`	160	446	70	26	0	Cruise Control Enable (Keypad B1)	Behind background (pill; label printed) · On #9aac7f dark / #859369 light · Off #3A3A40 dark / #E4E4E4 light
48	**LOG pill**	Indicator	`diagnostic_dense_indicator_fill_white.png`	236	446	77	26	0	Datalog Switch (Keypad B5)	Behind background (pill; label printed) · On #9aac7f dark / #859369 light · Off #3A3A40 dark / #E4E4E4 light
49	**CEL pill**	Indicator	`diagnostic_dense_indicator_fill_white.png`	319	446	114	26	0	Check Engine Light (DPO5)	Behind background (pill; label printed) · On #fbbf24 dark / #f59e0b light · Off #3A3A40 dark / #E4E4E4 light
50	**KILL pill**	Indicator	`diagnostic_dense_indicator_fill_white.png`	439	446	77	26	0	Kill Switch Input 1 (GenCond5)	Behind background (pill; label printed) · On #ef7676 dark / #b22222 light · Off #3A3A40 dark / #E4E4E4 light
51	**fuel value**	Numeric gauge		810	448	42	22	2	Fuel Level 1 (PD16 SPI4)	Value only · Outfit SemiBold 18px · right-aligned · #e4e4e7 dark · #18181b light
52	**trim value**	Numeric gauge		720	448	30	22	2	Rotary Trim Module 1 (Boost Trim)	Value only · Outfit SemiBold 18px · right-aligned · #e4e4e7 dark · #18181b light

PART 7

Empty versions: change what any slot shows

Every screen folder has an **Empty version** subfolder holding a background with only the hardware: chrome bezels, blank magnolia faces, data plates, the LCD and lower windows, or plain tiles. There are no labels, units, numbers, ticks, scales or lamps in the picture. NSP draws all of those, so a slot can show oil temp today and EGT next year with no new artwork.

It's the same build as Parts 5–6 with one difference: you **keep** the parts of each NSP component that the printed art used to cover, and style them to match.

1 Background

Image component, `…_background_…_empty.png`, X 0, Y 0, 860 × 480, Z 1. The empty backgrounds keep the same see-through bar slots as the printed ones (bottom of every tile and plate, the lower window, the long top tile), so bars go underneath at Z 0, exactly as in Part 5.

2 Dials on an empty face

1. Dial Gauge (retro circular) with its box set to the slot's box from the table. The box is the face, so ticks and numbers land on it.
2. Channel, Min/Max of your choice; Start −135°, End 135°. **Label Text**: what the gauge shows ("OIL", "EGT"…).
3. **Palette Colours**: ticks, numbers and label in the ink colour below, opacity 100%. Arc, background and centre at opacity 0. Use the warning colour for your redline or limit zone (Warning Min/Max).
4. Number font **Barlow Condensed SemiBold**; label font **Outfit Bold**. Choose major/minor tick counts so labels land on round numbers (0–100 → 5 majors; 0–8000 → 8 majors).
5. Needle: the same needle files as the printed screens (`…_needle_large` for the big faces, `…_needle_small` for 160 px faces, `sixpack_needle_medium` for the six-pack's 200 px faces), with the same −80% offset and a hub on top. If NSP draws the needle to the box size, it will now reach the tick ring by default.
6. If NSP's ticks sit too close to the chrome, shrink the box a few pixels on each side and keep the same centre.

3 Windows, plates and tiles

- **LCD window** (76 × 36): two value-only Numeric Gauges stacked (each 17 px tall), Barlow Condensed SemiBold 15 px, `#E9E04A` day / `#C9C03C` night. Or one larger value.
- **Lower window** (104 × 26): its bar slot (88 × 5) takes a bar from underneath. The top of the window is free for a small label.
- **Data plates** and **tiles**: one Numeric Gauge filling the slot with its **label** and **units** switched on. Plates: label Outfit Bold 12 px in ink, units Outfit SemiBold 10 px, value Barlow Condensed SemiBold ~54 px. Tiles: label Outfit SemiBold 11 px, units Outfit Medium 11 px, value Outfit SemiBold 38 px (12-tile) or 32 px (20-tile). Each one has a bar slot along the bottom: put a bar under it, or a plain bar at 0 so the slot shows the track colour.

4 Lamps and status lights anywhere

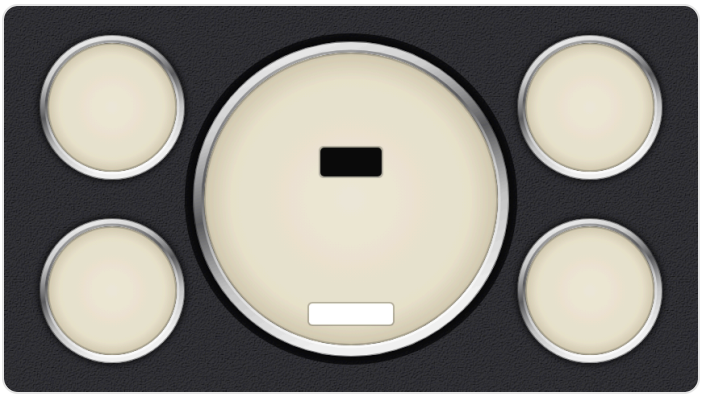
The empty backgrounds have no lamp or pill cut-outs, because where a lamp goes is up to you. To place one anywhere, put the Indicator **on top** of the background instead (Z 2) and give it both states. Use a …`_lamp_`…`_white.png` (classic screens) or …`_status_dot_white.png` (diagnostic) as the image. Set the **On** colour to the lamp colour and the **Off** colour to the "lamp off" colour below, so it reads as a dark lens until it lights. On the diagnostic screens, put a text label beside each dot in the status strip.

COLOUR ROLE	CLASSIC DAY	CLASSIC NIGHT	DIAGNOSTIC DARK	DIAGNOSTIC LIGHT
Ink (ticks, numbers, values)	#1D1B18	#E9E2CC	#E4E4E7	#18181B
Secondary (labels, units)	#3A362F	#B9B19B	#B6B6BC / units #9A9AA1	#4A4A4F / units #636369
Warning / redline	#ED7135	#F08A54	#F08A54	#ED7135
Danger	#B22222	#EF7676	#EF7676	#B22222
Bar fill	#1D1B18	#F08A54	#9AAC7F	#859369
Bar track (bar background colour)	#D6CDB4 windows · #CCC5B3 plates	#1D1C19 windows · #403D37 plates	#333337	#EBEBEB
Lamp off	#2A2620	#0D0C0B	#2A2A30	#E4E4E4
Needle	black (Day files)	orange (Night files)	—	—

Mixing the two. You can start from the printed screen and later rebuild just one gauge as "empty": keep the printed background, put a blank face image over that slot (crop it from the empty background), and add NSP ticks there. In practice it's quicker to switch the whole screen to the empty background once you start customising.

PART 7 · EMPTY LAYOUT

Five-gauge (screens A and B)



..._background_day_empty.png



..._background_night_empty.png

Each slot's box is where the component goes. For dials, the box is the face (centre = pivot). The last column says what the printed version puts there, if you want to start from that and change things.

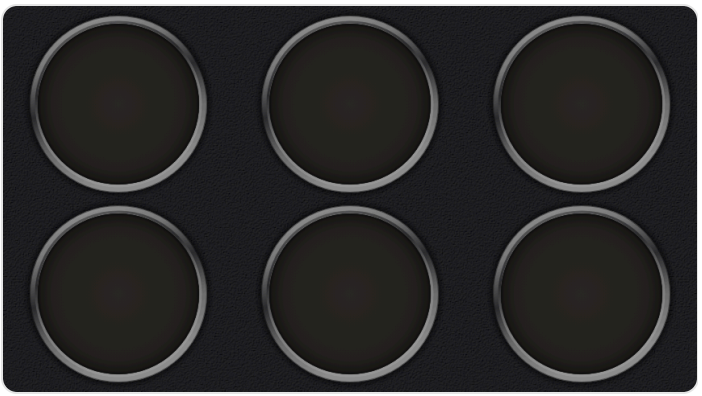
SLOT	KIND	X	Y	X SIZE	Y SIZE	CENTRE	IN THE PRINTED VERSION
Centre	Large dial	248	58	364	364	430, 240	A: Speedo 10–130 mph · B: Air/fuel 10–20 AFR
Centre · LCD window	Text window	392	176	76	36	430, 194	A: Odometer + trip · B: AFR + target
Centre · lower window	Bar / text window	378	370	104	26	430, 383	A: Fuel level bar · B: Throttle position bar
Centre · lower window bar	Bar (behind background)	386	383	88	5	430, 385	—
Top left	Dial	54	46	160	160	134, 126	A: Oil pressure 0–100 psi · B: Oil temp 50–150 °C
Bottom left	Dial	54	274	160	160	134, 354	A: Water 30–110 °C · B: Fuel pressure 0–80 psi
Top right	Dial	646	46	160	160	726, 126	A: Tacho 0–8000 rpm · B: Volts 8–16 V
Bottom right	Dial	646	274	160	160	726, 354	A: Boost −9.8–30 psi · B: Intake air 0–100 °C

PART 7 · EMPTY LAYOUT

Six-pack (screen C)



..._background_day_empty.png



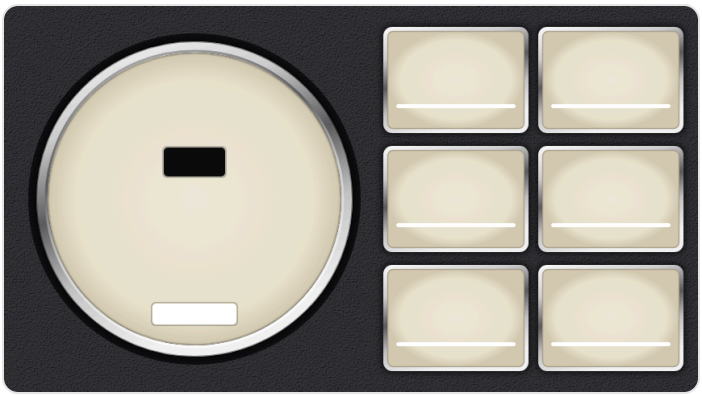
..._background_night_empty.png

Each slot's box is where the component goes. For dials, the box is the face (centre = pivot). The last column says what the printed version puts there, if you want to start from that and change things.

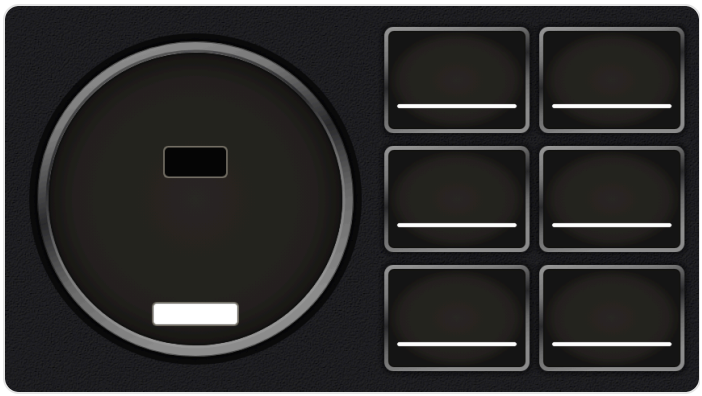
SLOT	KIND	X	Y	X SIZE	Y SIZE	CENTRE	IN THE PRINTED VERSION
Top left	Dial	43	22	200	200	143, 122	Air/fuel 10–20 AFR
Top centre	Dial	330	22	200	200	430, 122	Oil temp 50–150 °C
Top right	Dial	617	22	200	200	717, 122	Fuel pressure 0–80 psi
Bottom left	Dial	43	258	200	200	143, 358	Intake air 0–100 °C
Bottom centre	Dial	330	258	200	200	430, 358	Volts 8–16 V
Bottom right	Dial	617	258	200	200	717, 358	Ignition −10–50 °BTDC

PART 7 · EMPTY LAYOUT

Hybrid (screen D)



..._background_day_empty.png



..._background_night_empty.png

Each slot's box is where the component goes. For dials, the box is the face (centre = pivot). The last column says what the printed version puts there, if you want to start from that and change things.

SLOT	KIND	X	Y	X SIZE	Y SIZE	CENTRE	IN THE PRINTED VERSION
Left dial	Large dial	54	58	364	364	236, 240	Air/fuel 10–20 AFR (+ target needle)
Left dial · LCD window	Text window	198	176	76	36	236, 194	AFR + target
Left dial · lower window	Bar / text window	184	370	104	26	236, 383	Throttle position bar
Left dial · lower window bar	Bar (behind background)	192	383	88	5	236, 385	—
Plate 1	Data plate	475	31	170	122	560, 92	OIL TEMP 50–150 °C
Plate 1 · bar	Bar (behind background)	486	122	148	5	560, 124	—
Plate 2	Data plate	667	31	170	122	752, 92	FUEL PRESS 0–80 PSI
Plate 2 · bar	Bar (behind background)	678	122	148	5	752, 124	—
Plate 3	Data plate	475	179	170	122	560, 240	AIR TEMP 0–100 °C
Plate 3 · bar	Bar (behind background)	486	270	148	5	560, 272	—
Plate 4	Data plate	667	179	170	122	752, 240	BATTERY 8–16 VOLTS
Plate 4 · bar	Bar (behind background)	678	270	148	5	752, 272	—
Plate 5	Data plate	475	327	170	122	560, 388	IGNITION -10–50 °BTDC
Plate 5 · bar	Bar (behind background)	486	418	148	5	560, 420	—
Plate 6	Data plate	667	327	170	122	752, 388	INJ DUTY 0–100 %
Plate 6 · bar	Bar (behind background)	678	418	148	5	752, 420	—

PART 7 · EMPTY LAYOUT

Diagnostic, 12 tiles (screen E)



..._background_dark_empty.png

..._background_light_empty.png

Each slot's box is where the component goes. For dials, the box is the face (centre = pivot). The last column says what the printed version puts there, if you want to start from that and change things.

SLOT	KIND	X	Y	X SIZE	Y SIZE	CENTRE	IN THE PRINTED VERSION
Top 1 (wide-ish)	Tile	8	8	190	80	103, 48	RPM value
Top 2 (long, for a bar)	Tile	204	8	472	80	440, 48	RPM bar 0–8000
Top 3	Tile	682	8	170	80	767, 48	Speed (mph)
Tile 1 (row 1, col 1)	Tile	8	94	206	110	111, 149	BOOST -10–30 PSI
Tile 2 (row 1, col 2)	Tile	220	94	206	110	323, 149	AIR / FUEL 10–20 AFR
Tile 3 (row 1, col 3)	Tile	432	94	206	110	535, 149	IGNITION -10–50 °BTDC
Tile 4 (row 1, col 4)	Tile	644	94	206	110	747, 149	INJ DUTY 0–100 %
Tile 5 (row 2, col 1)	Tile	8	210	206	110	111, 265	COOLANT 30–120 °C
Tile 6 (row 2, col 2)	Tile	220	210	206	110	323, 265	OIL PRESS 0–100 PSI
Tile 7 (row 2, col 3)	Tile	432	210	206	110	535, 265	OIL TEMP 30–150 °C
Tile 8 (row 2, col 4)	Tile	644	210	206	110	747, 265	FUEL PRESS 0–80 PSI
Tile 9 (row 3, col 1)	Tile	8	326	206	110	111, 381	INTAKE AIR -10–80 °C
Tile 10 (row 3, col 2)	Tile	220	326	206	110	323, 381	BATTERY 10–16 V
Tile 11 (row 3, col 3)	Tile	432	326	206	110	535, 381	THROTTLE 0–100 %
Tile 12 (row 3, col 4)	Tile	644	326	206	110	747, 381	KNOCK 0–100 LVL
Top 2 · bar	Bar (behind background)	218	40	444	22	440, 51	—
Tile 1 · bar	Bar (behind background)	20	180	182	6	111, 183	—
Tile 2 · bar	Bar (behind background)	232	180	182	6	323, 183	—
Tile 3 · bar	Bar (behind background)	444	180	182	6	535, 183	—
Tile 4 · bar	Bar (behind background)	656	180	182	6	747, 183	—
Tile 5 · bar	Bar (behind background)	20	296	182	6	111, 299	—
Tile 6 · bar	Bar (behind background)	232	296	182	6	323, 299	—
Tile 7 · bar	Bar (behind background)	444	296	182	6	535, 299	—
Tile 8 · bar	Bar (behind background)	656	296	182	6	747, 299	—
Tile 9 · bar	Bar (behind background)	20	412	182	6	111, 415	—
Tile 10 · bar	Bar (behind background)	232	412	182	6	323, 415	—

SLOT	KIND	X	Y	X SIZE	Y SIZE	CENTRE	IN THE PRINTED VERSION
Tile 11 · bar	Bar (behind background)	444	412	182	6	535, 415	—
Tile 12 · bar	Bar (behind background)	656	412	182	6	747, 415	—
Status strip	Open area	8	444	844	30	430, 459	Status lamps + labels, fuel, boost trim

PART 7 · EMPTY LAYOUT

Diagnostic Dense, 20 tiles (screen F)



..._background_dark_empty.png

..._background_light_empty.png

Each slot's box is where the component goes. For dials, the box is the face (centre = pivot). The last column says what the printed version puts there, if you want to start from that and change things.

SLOT	KIND	X	Y	X SIZE	Y SIZE	CENTRE	IN THE PRINTED VERSION
Top 1	Tile	8	8	160	56	88, 36	RPM value
Top 2 (long, for a bar)	Tile	174	8	532	56	440, 36	RPM bar 0–8000
Top 3	Tile	712	8	140	56	782, 36	Speed (mph)
Tile 1 (row 1, col 1)	Tile	8	70	164	86	90, 113	BOOST -10–30 PSI
Tile 2 (row 1, col 2)	Tile	178	70	164	86	260, 113	BOOST TGT -10–30 PSI
Tile 3 (row 1, col 3)	Tile	348	70	164	86	430, 113	GEAR 0–5
Tile 4 (row 1, col 4)	Tile	518	70	164	86	600, 113	AIR / FUEL 10–20 AFR
Tile 5 (row 1, col 5)	Tile	688	70	164	86	770, 113	AFR TGT 10–20 AFR
Tile 6 (row 2, col 1)	Tile	8	162	164	86	90, 205	IGNITION -10–50 °BTDC
Tile 7 (row 2, col 2)	Tile	178	162	164	86	260, 205	KNOCK 0–100 LVL
Tile 8 (row 2, col 3)	Tile	348	162	164	86	430, 205	KNOCK RET 0–10 °
Tile 9 (row 2, col 4)	Tile	518	162	164	86	600, 205	INJ DUTY 0–100 %
Tile 10 (row 2, col 5)	Tile	688	162	164	86	770, 205	INJ PW 0–20 MS
Tile 11 (row 3, col 1)	Tile	8	254	164	86	90, 297	COOLANT 30–120 °C
Tile 12 (row 3, col 2)	Tile	178	254	164	86	260, 297	OIL PRESS 0–100 PSI
Tile 13 (row 3, col 3)	Tile	348	254	164	86	430, 297	OIL TEMP 30–150 °C
Tile 14 (row 3, col 4)	Tile	518	254	164	86	600, 297	FUEL PRESS 0–80 PSI
Tile 15 (row 3, col 5)	Tile	688	254	164	86	770, 297	INTAKE AIR -10–80 °C
Tile 16 (row 4, col 1)	Tile	8	346	164	86	90, 389	THROTTLE 0–100 %
Tile 17 (row 4, col 2)	Tile	178	346	164	86	260, 389	PEDAL 0–100 %
Tile 18 (row 4, col 3)	Tile	348	346	164	86	430, 389	BRAKE FRONT 0–1500 PSI
Tile 19 (row 4, col 4)	Tile	518	346	164	86	600, 389	BATTERY 10–16 V
Tile 20 (row 4, col 5)	Tile	688	346	164	86	770, 389	CL TRIM -25–25 %
Top 2 · bar	Bar (behind background)	186	29	508	12	440, 35	—
Tile 1 · bar	Bar (behind background)	18	142	144	4	90, 144	—
Tile 2 · bar	Bar (behind background)	188	142	144	4	260, 144	—

SLOT	KIND	X	Y	X SIZE	Y SIZE	CENTRE	IN THE PRINTED VERSION
Tile 3 · bar	Bar (behind background)	358	142	144	4	430, 144	—
Tile 4 · bar	Bar (behind background)	528	142	144	4	600, 144	—
Tile 5 · bar	Bar (behind background)	698	142	144	4	770, 144	—
Tile 6 · bar	Bar (behind background)	18	234	144	4	90, 236	—
Tile 7 · bar	Bar (behind background)	188	234	144	4	260, 236	—
Tile 8 · bar	Bar (behind background)	358	234	144	4	430, 236	—
Tile 9 · bar	Bar (behind background)	528	234	144	4	600, 236	—
Tile 10 · bar	Bar (behind background)	698	234	144	4	770, 236	—
Tile 11 · bar	Bar (behind background)	18	326	144	4	90, 328	—
Tile 12 · bar	Bar (behind background)	188	326	144	4	260, 328	—
Tile 13 · bar	Bar (behind background)	358	326	144	4	430, 328	—
Tile 14 · bar	Bar (behind background)	528	326	144	4	600, 328	—
Tile 15 · bar	Bar (behind background)	698	326	144	4	770, 328	—
Tile 16 · bar	Bar (behind background)	18	418	144	4	90, 420	—
Tile 17 · bar	Bar (behind background)	188	418	144	4	260, 420	—
Tile 18 · bar	Bar (behind background)	358	418	144	4	430, 420	—
Tile 19 · bar	Bar (behind background)	528	418	144	4	600, 420	—
Tile 20 · bar	Bar (behind background)	698	418	144	4	770, 420	—
Status strip	Open area	8	444	844	30	430, 459	Status lamps + labels, fuel, boost trim

PART 8

Write to the dash, day/night, troubleshooting

Order and write

1. In **Screens**, drag the screens into the order you want (up to five), and choose which one the dash starts on.
2. Set up the screen buttons if you haven't (one button cycling, or next/previous).
3. Write the dash map to the dash, then save the dash map to a new file.

Day and night

NSP's day/night setting switches colours, not background images. Two easy ways to use the night artwork: build the night version as its own screen (Classic · Night next to Classic · Day), or keep one screen and swap …_background_day for …_background_night and the needles for their _night versions (and the indicator off colours for their night values). The night needles are orange so they read on the dark faces.

Moving a finished screen into another dash map

From Haltech's video: open the dash map that holds the screen, add blank screens until it sits in the slot you want, then save. Open your main map and use **File › Import and upload**, pick that file, tick **import screens**, choose the screen and **import the selected items**. This adds screens without wiping the others.

If something looks off

- **Needle swings around the wrong point:** change Needle Image Offset (Part 5).
- **Needle lines up at one end but not the other:** check Min/Max and that Start/End are −135/135. The boost gauge is linear from −9.82 to 30 psi.
- **Numbers sit a few pixels off:** numeric gauges pad their text slightly differently by version. Nudge Y by eye with Simulation on.
- **A whole dial shows up behind our face:** a palette colour still has opacity above 0.
- **Everything is offset or scaled:** make sure the background image is X 0, Y 0, 860 × 480.
- **Starting warning colours** (set to your engine): coolant over 100 °C, oil under 15 psi above 2000 rpm, oil temp over 120 °C, battery under 12.5 V running, boost at your limit. Use NSP's Warning Min/Max so values turn orange or red.

Changing the artwork. Ranges, the 7000 rpm redline and labels are set in `4 - Reference/Generator source`. Change a value and re-run to make new PNGs at the same positions.

Classic Mini DIY by Cole Gentry. Fonts: Outfit and Barlow Condensed, SIL Open Font License. Method references: Haltech, "How To Customise Your Haltech Dash" and "Tips and Tricks: How to Import Custom Screens and Needles" (YouTube).